

ROTARY TABLE TECHNICAL DOCUMENTATION

IDENTITY INFORMATION

Company_Name: Permian Basin Drilling Solutions LLC

Field_Engineer: Marcus J. Hendricks

Field_Engineer_Id: PBD-ENG-4782

Document_Version: 2.3.1

Machine_Name: Primary Wellhead Rotary Assembly Unit 7

Machine_Id: PBDS-RTA-2847-07

Installation_Date: 2022-03-15

Number_Of_Units_In_Machine: 3

UNIT 1: MAIN ROTARY TABLE DRIVE

Unit_Name: Primary Rotary Table Drive Assembly

Unit_Id: PBDS-RTD-2847-01

Manufacture_Date: 2021-11-22

Manufacturer_Company_Name: Weatherford International Solutions

Manufacturer_Company_Id: WIS-MFG-7734

Inventory_Quantity: 2 units in stock

Supplier_Information: Weatherford Houston Distribution Center, Lead Time: 14 days

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** 15,000 feet (4,572 meters)

- **Operating_Pressure_PSI:** 5,000 PSI maximum continuous

- **Drilling_Fluid_Compatibility:** Water-based, synthetic, and oil-based muds (API RP 13B-1 compliant)

- **Known_Error_Codes:** RT-E001 (bearing wear), RT-E003 (seal degradation), RT-E007 (torque loss >15%)

- **Replacement_Part_Number:** WIS-RTD-4400-A2

- **Maintenance_Schedule:** 500-hour intervals; full bearing inspection every 2,000 hours

SECTION 1 - USE CASE

- Primary rotational drive mechanism for drill string rotation during wellbore drilling operations

- Transmits torque from kelly drive system to drill pipe at controlled RPM (0-200 RPM range)

- Supports vertical load capacity of 750,000 lbs during static drilling conditions

- Enables directional drilling control through precise rotational speed modulation
- Integrates with automated drilling control systems for real-time torque monitoring

SECTION 2 - FAILURE CAUSES

- **RT-E001 (Bearing Wear):** Frequency 35% — Inadequate lubrication intervals; contaminated drilling fluid ingress
- **RT-E003 (Seal Degradation):** Frequency 28% — Thermal cycling stress; abrasive particle erosion in mud system
- **RT-E007 (Torque Loss):** Frequency 18% — Slippage in drive coupling; worn bushing surfaces
- **RT-E002 (Vibration Anomaly):** Frequency 12% — Misalignment with kelly drive; unbalanced drill string
- **RT-E005 (Pressure Spike):** Frequency 7% — Sudden mud viscosity increase; blockage in circulation lines

SECTION 3 - SOLUTIONS

- **RT-E001 Response:** Increase lubrication frequency to 250-hour intervals; replace bearing cartridge per WIS-RTD-4400-A2 specifications; inspect seals during service (OSHA 1910.147 lockout/tagout required)
- **RT-E003 Response:** Install upgraded elastomer seals rated for 180°F continuous operation; implement mud filtration system maintaining <4 micron particle size per API RP 13B-1
- **RT-E007 Response:** Realign drive coupling within 0.005-inch tolerance; replace worn bushings; verify torque transmission at 25%, 50%, 75%, and 100% load increments
- **RT-E002 Response:** Perform dynamic balancing analysis; check kelly drive concentricity; reduce RPM by 10% until vibration signature normalizes
- **RT-E005 Response:** Flush circulation system; verify mud rheology compliance; install pressure relief valve set at 5,200 PSI per API RP 54 standards

SECTION 4 - INVENTORY DATA

- **Current Stock:** 2 complete drive assemblies; 8 bearing cartridges; 12 seal kits in warehouse
- **Reorder Threshold:** Maintain minimum 1 complete unit + consumables; trigger reorder at 25% stock level
- **Primary Supplier:** Weatherford Houston Distribution Center; Secondary: Smith Services Equipment (lead time 21 days)
- **Lead Time & Cost:** 14 days standard; expedited 5-day option available at 18% premium; unit cost \$47,500 USD

UNIT 2: ROTARY TABLE BEARING ASSEMBLY

Unit_Name: Thrust & Radial Bearing Stack Assembly

Unit_Id: PBDS-RTB-2847-02

Manufacture_Date: 2021-10-18

Manufacturer_Company_Name: SKF Industrial Bearings North America

****Manufacturer_Company_Id:** SKF-NA-5521**

****Inventory_Quantity:** 4 cartridge sets in stock**

****Supplier_Information:** SKF Dallas Regional Warehouse, Lead Time: 7 days**

DOMAIN SPECIFICATIONS

- ****Wellbore_Depth_Rating:** 15,000 feet (compatible with primary drive)**
- ****Operating_Pressure_PSI:** 3,500 PSI radial load equivalent**
- ****Drilling_Fluid_Compatibility:** Sealed design; compatible with all mud types per API RP 13B-1**
- ****Known_Error_Codes:** RT-B001 (preload loss), RT-B004 (cage wear), RT-B006 (spalling initiation)**
- ****Replacement_Part_Number:** SKF-RTB-6200-HC**
- ****Maintenance_Schedule:** Inspect every 500 hours; replace cartridge every 3,000 operating hours or upon RT-E001 detection**

SECTION 1 - USE CASE

- Supports combined thrust and radial loads from rotating drill string assembly
- Maintains rotational alignment within 0.002-inch runout tolerance during full-load drilling
- Provides low-friction bearing surface enabling smooth RPM transitions (0-200 RPM)
- Isolates drilling fluid contamination through dual-seal design
- Enables rapid cartridge replacement without full assembly disassembly

SECTION 2 - FAILURE CAUSES

- ****RT-B001 (Preload Loss):** Frequency 42% — Thermal expansion/contraction cycles; inadequate bolt torque maintenance**
- ****RT-B004 (Cage Wear):** Frequency 31% — Bearing skidding under variable load conditions; insufficient lubrication film**
- ****RT-B006 (Spalling Initiation):** Frequency 18% — Subsurface fatigue from cyclic loading; contaminated lubricant**
- ****RT-B003 (Seal Failure):** Frequency 6% — Mud ingress past primary seal; thermal degradation of elastomer**
- ****RT-B005 (Noise/Vibration):** Frequency 3% — Bearing race defects; loose mounting hardware**

SECTION 3 - SOLUTIONS

- ****RT-B001 Response:** Verify preload bolt torque at 450 ft-lbs $\pm$ 10 ft-lbs per SKF specifications; use calibrated torque wrench; document torque values in maintenance log per OSHA 1910.176**
- ****RT-B004 Response:** Replace bearing cartridge immediately upon detection; upgrade to SKF-RTB-6200-HC with reinforced cage design; increase lubrication frequency to 250-hour intervals**
- ****RT-B006 Response:** Implement condition monitoring (vibration analysis) at 500-hour intervals; replace cartridge before spalling propagates; flush bearing cavity with ISO VG 32 turbine oil**
- ****RT-B003 Response:** Inspect dual seals during each 500-hour service; replace if any mud contamination detected; verify seal installation depth per SKF technical manual**

- **RT-B005 Response:** Tighten all mounting bolts to specification; perform dynamic balancing; reduce load by 15% until noise signature resolves

SECTION 4 - INVENTORY DATA

- **Current Stock:** 4 complete bearing cartridge sets; 16 individual seal rings; 8 preload springs
- **Reorder Threshold:** Maintain minimum 2 cartridge sets; reorder when stock drops below 3 units
- **Primary Supplier:** SKF Dallas Regional Warehouse; Secondary: Timken Industrial Distribution (lead time 10 days)
- **Lead Time & Cost:** 7 days standard delivery; cartridge cost \$8,200 USD; seal kit \$1,100 USD per set

UNIT 3: ROTARY TABLE DRIVE COUPLING & TRANSMISSION

Unit_Name: Kelly Drive to Rotary Table Coupling Assembly

Unit_Id: PBDS-RTC-2847-03

Manufacture_Date: 2021-09-30

Manufacturer_Company_Name: Rexnord Industrial Transmission Systems

Manufacturer_Company_Id: RIX-MFG-3344

Inventory_Quantity: 1 complete coupling assembly in stock

Supplier_Information: Rexnord Houston Service Center, Lead Time: 10 days

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** 15,000 feet (integrated system rating)
- **Operating_Pressure_PSI:** 6,000 PSI equivalent torque load (750 ft-lbs continuous)
- **Drilling_Fluid_Compatibility:** Sealed coupling; mud splash zone rated for all fluid types
- **Known_Error_Codes:** RT-C002 (elastomer degradation), RT-C004 (misalignment), RT-C008 (slip ring wear)
- **Replacement_Part_Number:** RIX-RTC-7750-B1
- **Maintenance_Schedule:** Alignment check every 250 hours; elastomer inspection every 500 hours; full replacement every 5,000 hours

SECTION 1 - USE CASE

- Transmits rotational torque from kelly drive system to primary rotary table drive assembly
- Accommodates minor angular and axial misalignment (± 2 degrees, ± 0.25 inches) without load loss
- Provides vibration damping through elastomer element reducing stress on bearing assembly
- Enables rapid disconnect for maintenance operations per OSHA 1910.147 lockout/tagout procedures
- Integrates torque monitoring sensors for automated drilling control feedback

SECTION 2 - FAILURE CAUSES

- **RT-C002 (Elastomer Degradation):** Frequency 38% — Thermal cycling (ambient to 140°F); ozone exposure in mud pit environment
- **RT-C004 (Misalignment):** Frequency 29% — Settling of drilling rig substructure; kelly drive drift over time
- **RT-C008 (Slip Ring Wear):** Frequency 18% — Inadequate lubrication of slip interface; abrasive mud particle contamination
- **RT-C001 (Torque Loss):** Frequency 10% — Elastomer shear failure; coupling bolt loosening
- **RT-C006 (Noise Generation):** Frequency 5% — Bearing wear in coupling hub; elastomer cracking

SECTION 3 - SOLUTIONS

- **RT-C002 Response:** Replace elastomer element per RIX-RTC-7750-B1 specifications every 5,000 hours or upon visual cracking; store replacement elements in climate-controlled warehouse (60-75°F); apply UV-protective coating to exposed elastomer
- **RT-C004 Response:** Perform laser alignment check at 250-hour intervals; realign kelly drive to rotary table within 0.005-inch concentricity tolerance; document alignment data per API RP 54; use dial indicators per OSHA 1910.212 machinery guarding standards
- **RT-C008 Response:** Lubricate slip ring interface with ISO VG 32 turbine oil every 250 hours; inspect for mud ingress; replace slip ring assembly if wear exceeds 0.010-inch radial clearance
- **RT-C001 Response:** Verify all coupling bolts at 350 ft-lbs torque per Rexnord specifications; use thread-locking compound (Loctite 243) on all fasteners; perform torque verification every 500 hours
- **RT-C006 Response:** Inspect coupling hub bearings during 500-hour service; replace if noise persists after lubrication; reduce RPM by 20% until replacement completed

SECTION 4 - INVENTORY DATA

- **Current Stock:** 1 complete coupling assembly; 3 elastomer element kits; 2 slip ring cartridges
- **Reorder Threshold:** Maintain minimum 1 complete unit; trigger reorder at 50% consumable stock level
- **Primary Supplier:** Rexnord Houston Service Center; Secondary: Applied Industrial Technologies (lead time 12 days)
- **Lead Time & Cost:** 10 days standard; coupling assembly \$34,800 USD; elastomer kit \$5,200 USD; expedited 3-day option at 22% premium

COMPLIANCE & SAFETY NOTES

OSHA 1910 Compliance:

- All maintenance procedures require lockout/tagout (LOTO) per 1910.147 before bearing or coupling service
- Personnel must wear hard hats, safety glasses, and steel-toed boots during all operations (1910.95, 1910.133)
- Rotating equipment guarding per 1910.212 must remain in place during operation

API RP 54 Alignment:

- All torque specifications and maintenance intervals comply with API Recommended Practice 54 (Recommended Practices for Occupational Safety and Health Program Management in the Petroleum and Natural Gas Industries)
- Mud system compatibility verified per API RP 13B-1 (Recommended Practice for Field Testing Water-Based Drilling Fluids)
- Bearing preload and alignment procedures follow API RP 54 equipment maintenance standards

****Document Approval:****

- Field Engineer: Marcus J. Hendricks (PBD-ENG-4782)
- Reviewed: 2024-01-10
- Next Review Date: 2025-01-10

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